

[2026-1] Machine Learning Term Project

Deadline: 11:59 PM, June 1st, 2026

1. Overview

This term project asks you to apply the machine learning concepts and techniques to a real-world problem of your own choosing. You will define a problem you find meaningful, select an appropriate open dataset, train and evaluate one or more machine learning models, and analyze the results.

The goal of this project is not simply to "run a model" — it is to gain hands-on experience with the entire ML pipeline, from problem formulation to result interpretation.

2. Project Procedure

2.1 Defining the Problem (Task)

Choose a concrete problem you would like to solve in a domain that interests you. Rather than stating "I will solve a classification problem," clearly explain why the problem matters, who would benefit, and what value the solution provides.

- **Example 1:** Automatically detect pneumonia from chest X-ray images to support clinicians' initial screening.
- **Example 2:** Predict star ratings from online shopping reviews to estimate sentiment for reviews where the rating is missing.
- **Example 3:** Forecast hourly demand for a public bike-sharing system based on time and weather conditions.

2.2 Selecting a Dataset

Choose an open dataset that fits your problem. The following sources are good starting points:

- Kaggle Datasets (<https://www.kaggle.com/datasets>)
- Hugging Face Datasets (<https://huggingface.co/datasets>)
- Google Dataset Search (<https://datasetsearch.research.google.com/>)
- AI Hub (<https://www.aihub.or.kr/>)

Please consider the following when choosing your dataset:

- Verify that the license permits academic use
- Avoid datasets that are too small (e.g., a few dozen samples) or impractically large (e.g., billions of samples)
- Projects that rely solely on well-known toy datasets (e.g., MNIST, CIFAR-10, ImageNet) may receive limited credit for novelty.

2.3 Modeling and Experiments

You may use any models of your choice (either the classical machine learning ones covered in class or advanced deep learning models), as long as your project meets the following requirements:

- Establish at least one simple baseline for comparison
- Build the model of your choice and choose appropriate hyperparameters by tuning them
- Measuring at least two evaluation metrics appropriate for your problem (e.g., Accuracy, F1-score, ROC-AUC, RMSE)
- Apply a clear train/validation/test split or cross-validation procedure

2.4 Result Analysis and Discussion

Summarize your experimental results in tables and figures, and discuss the following:

- Which model performed best, and why do you think it did?
- What characterizes the cases where the model failed (misclassifications, large errors)? You may suggest some qualitative results of the failure mode.
- Which aspects of the data make the learning task difficult?
- Is the result satisfactory for deployment? Given more time and resources, what would you try next?

3. Deliverables

- **Final report (PDF):** problem definition, dataset description, methodology, experiments, result analysis, conclusion, and references
- **Presentation slides (PDF):** suited for 5-minute presentation
- **Source code (GitHub repository link):** include README, requirements.txt or environment.yaml, and clear reproduction instructions

4. Important Notes

- There are no restrictions on the choice of deep learning libraries (e.g., Tensorflow, Pytorch).
- The presentation slides should be self-contained; assume your audience has no background in both machine learning and the targeted domain.
- The final report should include all the details of the project.
- Use of generative AI tools (ChatGPT, Claude, etc.) is permitted, but the scope and manner of use must be transparently documented in the report.
- If you reference others' code, you must cite the source; verbatim copying will be treated as plagiarism.
- Simply reproducing top Kaggle solutions on the same dataset will not receive credit.